Introduction to Nonconvex Optimization

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Roadmap

- Motivation: where nonconvexity appears and what guarantees we seek
- Part I: a concrete running example (rank-1 matrix factorization)
- Part II: generic local convergence tools (strong convexity, smoothness, RC)
- Part III: return to the example for local and global geometry

Where Nonconvex Optimization Appears

- Low-rank estimation: matrix factorization, PCA, matrix sensing
- Inverse problems: phase retrieval, blind deconvolution
- Representation learning: dictionary learning, tensor decomposition
- Deep learning: training neural networks

Different applications, but recurring questions: geometry, initialization, and algorithmic guarantees.

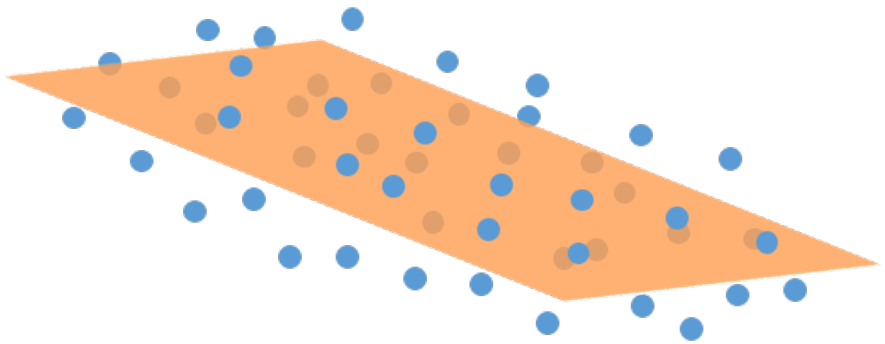
What Guarantees Do We Usually Seek?

- First-order guarantee: find $\boldsymbol{x}$ with $\|\nabla f(\boldsymbol{x})\|$ small
- Second-order guarantee: avoid strict saddles and reach local minima
- Global guarantee: reach a global minimizer (often under benign geometry)

- Typical intro hierarchy: stationary point $\rightarrow$ local minimum $\rightarrow$ global optimum
- Assumptions become stronger as guarantees get stronger
- In this lecture: understand when simple methods are still effective

A running example: rank-1 matrix factorization

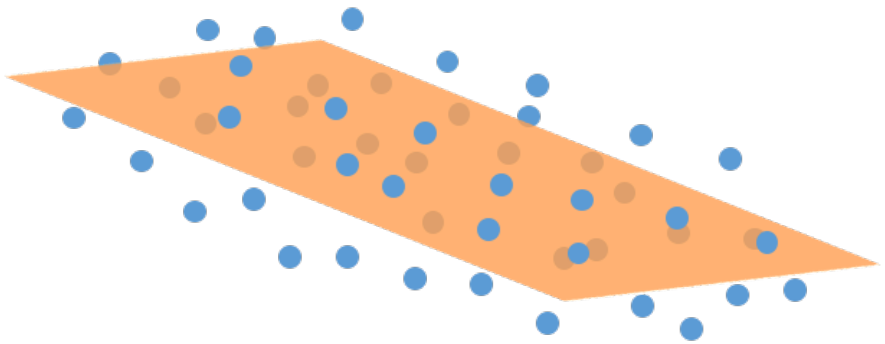
Principal Component Analysis



Given $\boldsymbol{M} \succeq \boldsymbol{0} \in \mathbb{R}^{n \times n}$ (not necessarily low-rank), find its best rank-r approximation:

$$\underbrace{\widehat{\boldsymbol{M}} = \arg\min_{\boldsymbol{Z}} \ \|\boldsymbol{Z} - \boldsymbol{M}\|_{\mathrm{F}}^2 \quad \text{s.t.} \quad \mathrm{rank}(\boldsymbol{Z}) \leq r}_{\text{nonconvex optimization!}}$$

Principal Component Analysis



This problem admits a closed-form solution.

- Let $\boldsymbol{M} = \sum_{i=1}^{n} \lambda_i \boldsymbol{u}_i \boldsymbol{u}_i^\top$ be eigen-decomposition of $\boldsymbol{M}$ ($\lambda_1 \geq \cdots \lambda_r > \lambda_{r+1} \geq \lambda_n$), then

$$\widehat{\boldsymbol{M}} = \sum_{i=1}^{r} \lambda_i \boldsymbol{u}_i \boldsymbol{u}_i^\top$$

— *nonconvex, but tractable*

Optimization Viewpoint

If we factorize $\boldsymbol{Z} = \boldsymbol{X}\boldsymbol{X}^\top$ with $\boldsymbol{X} \in \mathbb{R}^{n \times r}$, then it leads to a nonconvex problem:

$$\text{minimize}_{\boldsymbol{X} \in \mathbb{R}^{n \times r}} \quad f(\boldsymbol{X}) = \frac{1}{4}\|\boldsymbol{X}\boldsymbol{X}^\top - \boldsymbol{M}\|_{\mathrm{F}}^2$$

To simplify exposition, set $r = 1$:

$$\text{minimize}_{\boldsymbol{x}} \quad f(\boldsymbol{x}) = \frac{1}{4}\|\boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{M}\|_{\mathrm{F}}^2$$

Why Is the Global Minimizer the Leading Eigenvector? (I)

For the rank-1 objective

$$f(\boldsymbol{x}) = \frac{1}{4}\|\boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{M}\|_{\mathrm{F}}^2,$$

expand the Frobenius norm:

$$\begin{aligned} 4f(\boldsymbol{x}) &= \|\boldsymbol{x}\boldsymbol{x}^\top\|_{\mathrm{F}}^2 - 2\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{M}\rangle + \|\boldsymbol{M}\|_{\mathrm{F}}^2 \\ &= \|\boldsymbol{x}\|_2^4 - 2\boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x} + \|\boldsymbol{M}\|_{\mathrm{F}}^2. \end{aligned}$$

Let $\boldsymbol{x} = r\boldsymbol{v}$ with $\|\boldsymbol{v}\|_2 = 1$. Then

$$4f(r\boldsymbol{v}) = r^4 - 2r^2\boldsymbol{v}^\top \boldsymbol{M}\boldsymbol{v} + \|\boldsymbol{M}\|_{\mathrm{F}}^2.$$

For fixed r, minimizing f means maximizing $\boldsymbol{v}^\top \boldsymbol{M}\boldsymbol{v}$.

Why Is the Global Minimizer the Leading Eigenvector? (II)

By the Rayleigh quotient bound for $\boldsymbol{M} \succeq \boldsymbol{0}$,

$$\boldsymbol{v}^\top \boldsymbol{M} \boldsymbol{v} \leq \lambda_1,$$

with equality iff $\boldsymbol{v}$ is a leading eigenvector $\boldsymbol{u}_1$.

Thus the best direction is $\boldsymbol{v} = \pm \boldsymbol{u}_1$. Substituting gives

$$4f(r\boldsymbol{u}_1) = r^4 - 2\lambda_1 r^2 + \|\boldsymbol{M}\|_{\mathrm{F}}^2,$$

whose minimum is attained at $r^2 = \lambda_1$.

Therefore the global minimizers are

$$\boldsymbol{x}_{\mathsf{opt}} = \pm\sqrt{\lambda_1}\, \boldsymbol{u}_1$$

(unique up to sign when $\lambda_1 > \lambda_2$).

Part I Takeaway

- Nonconvex objectives can still have transparent global structure
- In this example, the global minimizer is explicit: $\boldsymbol{x}_{\mathsf{opt}} = \pm\sqrt{\lambda_1}\boldsymbol{u}_1$

- Remaining question: when do iterative algorithms provably reach such solutions?
- Next: build a generic toolkit to answer this

Part II: Generic Nonconvex Toolkit

Local geometry + algorithmic guarantees

Unconstrained Optimization

Consider an unconstrained optimization problem

$$\text{minimize}_{\boldsymbol{x}} \qquad f(\boldsymbol{x})$$

- For simplicity, we assume $f(\boldsymbol{x})$ is twice differentiable
- We assume the minimizer $\boldsymbol{x}_{\text{opt}}$ exists, i.e.,

$$\boldsymbol{x}_{\text{opt}} := \arg\min_{\boldsymbol{x}} f(\boldsymbol{x})$$

Critical/Stationary Points

Definition 1

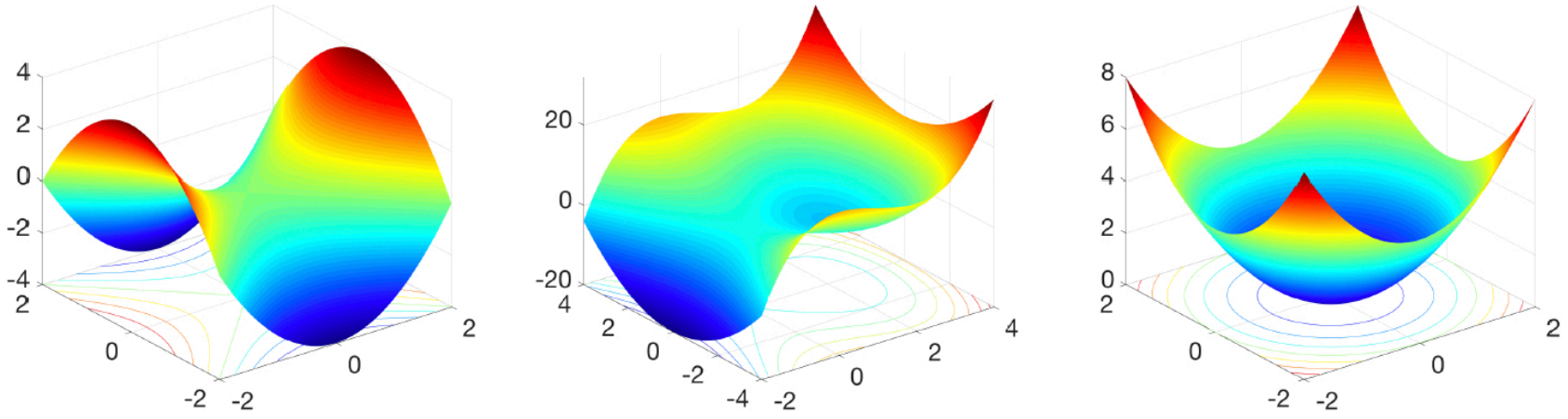
A first-order critical point of f satisfies

$$\nabla f(\boldsymbol{x}) = \mathbf{0}$$

- If f is convex, any 1st-order critical point is a global minimizer
- Finding 1st-order stationary point is sufficient for convex optimization
- Example: gradient descent (GD)

What Changes in Nonconvex Optimization?

First-order critical points could be global min, local min, local max, saddle points...



(a) strict saddle (b) local minimum (c) global minimum

figure credit: Li et al. '16

Simple algorithms like GD can get stuck at undesired stationary points.

Types of Critical Points

Definition 2

A second-order critical point $\boldsymbol{x}$ satisfies

$$\nabla f(\boldsymbol{x}) = \mathbf{0} \quad \text{and} \quad \nabla^2 f(\boldsymbol{x}) \succeq \mathbf{0}$$

For any first-order critical point $\boldsymbol{x}$:

- $\nabla^2 f(\boldsymbol{x}) \prec \mathbf{0}$ $\rightarrow$ local maximum
- $\nabla^2 f(\boldsymbol{x}) \succ \mathbf{0}$ $\rightarrow$ local minimum
- $\lambda_{\min}(\nabla^2 f(\boldsymbol{x})) < 0$ $\rightarrow$ *strict* saddle point

What Structure Makes Nonconvex Problems Solvable?

- Good local curvature near the target (strong convexity + smoothness)
- Benign global geometry (no spurious minima / strict saddles)

Local Strong Convexity and Smoothness

Definition 3

A twice differentiable function $f : \mathbb{R}^n \mapsto \mathbb{R}$ is said to be α-strongly convex in a set $\mathcal{B}$ if for all $\boldsymbol{x} \in \mathcal{B}$

$$\nabla^2 f(\boldsymbol{x}) \succeq \alpha \boldsymbol{I}_n.$$

Definition 4

A twice differentiable function $f : \mathbb{R}^n \mapsto \mathbb{R}$ is said to be β-smooth in a set $\mathcal{B}$ if for all $\boldsymbol{x} \in \mathcal{B}$

$$\|\nabla^2 f(\boldsymbol{x})\| \leq \beta.$$

Gradient Descent Theory Revisited

Gradient descent method with step size $\eta > 0$

$$\boldsymbol{x}^{t+1} = \boldsymbol{x}^t - \eta \nabla f(\boldsymbol{x}^t)$$

Lemma 5

Suppose f is α-strongly convex and β-smooth in the local ball $\mathcal{B}_\delta(\boldsymbol{x}_{\text{opt}}) := \{\boldsymbol{x} \mid \|\boldsymbol{x} - \boldsymbol{x}_{\text{opt}}\|_2 \le \delta\}$. Running gradient descent from $\boldsymbol{x}^0 \in \mathcal{B}_\delta(\boldsymbol{x}_{\text{opt}})$ with $\eta = 1/\beta$ achieves linear convergence

$$\|\boldsymbol{x}^t - \boldsymbol{x}_{\text{opt}}\|_2 \le \left(1 - \frac{\alpha}{\beta}\right)^t \|\boldsymbol{x}^0 - \boldsymbol{x}_{\text{opt}}\|_2, \quad t = 0, 1, 2, \ldots$$

Implications

- Condition number β/α determines rate of convergence
- Attains ε-accuracy (i.e., $\|\boldsymbol{x}^t - \boldsymbol{x}_{\text{opt}}\|_2 \le \varepsilon \|\boldsymbol{x}_{\text{opt}}\|_2$) within

$$O\left(\frac{\beta}{\alpha}\log\frac{1}{\varepsilon}\right)$$

  iterations
- Needs initialization $\boldsymbol{x}^0 \in \mathcal{B}_\delta(\boldsymbol{x}_{\text{opt}})$: basin of attraction

Proof of Lemma 5

Since $\nabla f(\boldsymbol{x}_{\mathsf{opt}}) = \boldsymbol{0}$, we can rewrite GD as

$$\begin{aligned}\boldsymbol{x}^{t+1} - \boldsymbol{x}_{\mathsf{opt}} &= \boldsymbol{x}^t - \eta \nabla f(\boldsymbol{x}^t) - [\boldsymbol{x}_{\mathsf{opt}} - \eta \nabla f(\boldsymbol{x}_{\mathsf{opt}})] \\ &= \left[\boldsymbol{I}_n - \eta \int_0^1 \nabla^2 f(\boldsymbol{x}(\tau)) \mathrm{d}\tau\right] (\boldsymbol{x}^t - \boldsymbol{x}_{\mathsf{opt}}),\end{aligned}$$

where $\boldsymbol{x}(\tau) := \boldsymbol{x}_{\mathsf{opt}} + \tau(\boldsymbol{x}^t - \boldsymbol{x}_{\mathsf{opt}})$. By local strong convexity and smoothness, one has

$$\alpha \boldsymbol{I}_n \preceq \nabla^2 f(\boldsymbol{x}(\tau)) \preceq \beta \boldsymbol{I}_n, \qquad \text{for all } 0 \le \tau \le 1$$

Therefore $\eta = 1/\beta$ yields

$$\boldsymbol{0} \preceq \boldsymbol{I}_n - \eta \int_0^1 \nabla^2 f(\boldsymbol{x}(\tau)) \mathrm{d}\tau \preceq (1 - \frac{\alpha}{\beta}) \boldsymbol{I}_n,$$

which further implies

$$\|\boldsymbol{x}^{t+1} - \boldsymbol{x}_{\mathsf{opt}}\|_2 \le \left(1 - \frac{\alpha}{\beta}\right) \|\boldsymbol{x}^t - \boldsymbol{x}_{\mathsf{opt}}\|_2$$

Regularity Condition

More generally, for update rule

$$\boldsymbol{x}^{t+1} = \boldsymbol{x}^t - \eta \boldsymbol{g}(\boldsymbol{x}^t),$$

where $\boldsymbol{g}(\cdot) : \mathbb{R}^n \mapsto \mathbb{R}^n$

Definition 6

$\boldsymbol{g}(\cdot)$ is said to obey RC(μ, λ, δ) for some $\mu, \lambda, \delta > 0$ if

$$2\langle \boldsymbol{g}(\boldsymbol{x}), \boldsymbol{x} - \boldsymbol{x}_{\text{opt}} \rangle \geq \mu \|\boldsymbol{g}(\boldsymbol{x})\|_2^2 + \lambda \|\boldsymbol{x} - \boldsymbol{x}_{\text{opt}}\|_2^2 \quad \forall \boldsymbol{x} \in \mathcal{B}_\delta(\boldsymbol{x}_{\text{opt}})$$

- Negative search direction $\boldsymbol{g}$ is positively correlated with error $\boldsymbol{x} - \boldsymbol{x}_{\text{opt}} \implies$ one-step improvement
- $\mu\lambda \leq 1$ by Cauchy-Schwarz

RC = One-Point Strong Convexity + Smoothness

- One-point α-strong convexity:

$$f(\boldsymbol{x}_{\mathsf{opt}}) - f(\boldsymbol{x}) \geq \langle \nabla f(\boldsymbol{x}), \boldsymbol{x}_{\mathsf{opt}} - \boldsymbol{x} \rangle + \frac{\alpha}{2} \|\boldsymbol{x} - \boldsymbol{x}_{\mathsf{opt}}\|_2^2 \tag{1}$$

- β-smoothness:

$$\begin{aligned} f(\boldsymbol{x}_{\mathsf{opt}}) - f(\boldsymbol{x}) &\leq f\Big(\boldsymbol{x} - \frac{1}{\beta}\nabla f(\boldsymbol{x})\Big) - f(\boldsymbol{x}) \\ &\leq \Big\langle \nabla f(\boldsymbol{x}), -\frac{1}{\beta}\nabla f(\boldsymbol{x}) \Big\rangle + \frac{\beta}{2}\Big\|\frac{1}{\beta}\nabla f(\boldsymbol{x})\Big\|_2^2 \\ &= -\frac{1}{2\beta}\,\|\nabla f(\boldsymbol{x})\|_2^2 \end{aligned} \tag{2}$$

RC = One-Point Strong Convexity + Smoothness

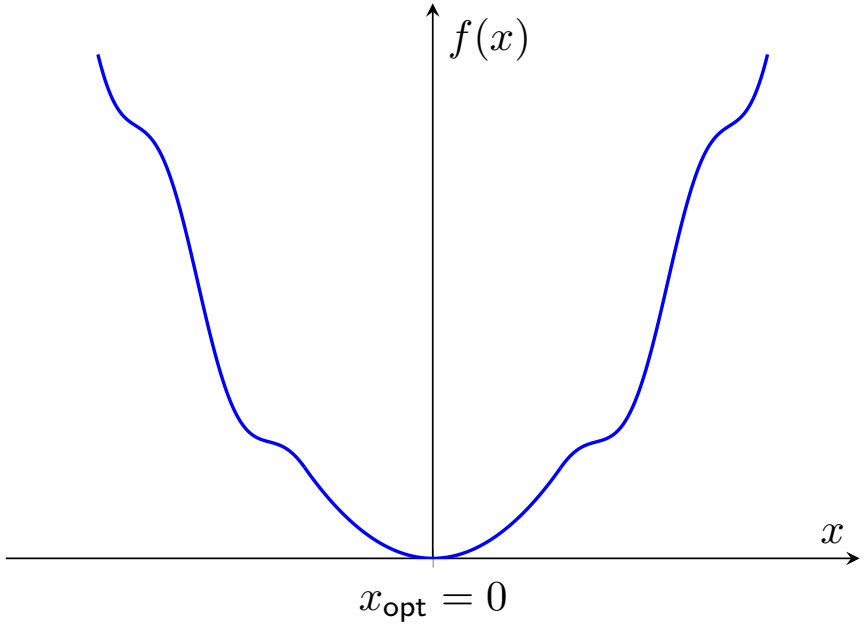
Combining relations (1) and (2) yields

$$\langle \nabla f(\boldsymbol{x}), \boldsymbol{x} - \boldsymbol{x}_{\mathsf{opt}} \rangle \geq \frac{\alpha}{2} \|\boldsymbol{x} - \boldsymbol{x}_{\mathsf{opt}}\|_2^2 + \frac{1}{2\beta} \|\nabla f(\boldsymbol{x})\|_2^2$$

— *RC holds with* $\mu = 1/\beta$ *and* $\lambda = \alpha$

Example of Nonconvex Functions

When $\boldsymbol{g}(\boldsymbol{x}) = \nabla f(\boldsymbol{x})$, f is not necessarily convex



$$f(x) = \begin{cases} x^2, & |x| \leq 6, \\ x^2 + 1.5|x|(\cos(|x| - 6) - 1), & |x| > 6 \end{cases}$$

Convergence under RC

Lemma 7

Suppose $\boldsymbol{g}(\cdot)$ *obeys* $\mathsf{RC}(\mu, \lambda, \delta)$. *The update rule* ($\boldsymbol{x}^{t+1} = \boldsymbol{x}^t - \eta \boldsymbol{g}(\boldsymbol{x}^t)$) *with* $\eta = \mu$ *and* $\boldsymbol{x}^0 \in \mathcal{B}_\delta(\boldsymbol{x}_{\mathsf{opt}})$ *obeys*

$$\|\boldsymbol{x}^t - \boldsymbol{x}_{\mathsf{opt}}\|_2^2 \le (1 - \mu\lambda)^t \, \|\boldsymbol{x}^0 - \boldsymbol{x}_{\mathsf{opt}}\|_2^2$$

- $\boldsymbol{g}(\cdot)$: more general search directions
  - example: in vanilla GD, $\boldsymbol{g}(\boldsymbol{x}) = \nabla f(\boldsymbol{x})$
- The product $\mu\lambda$ determines the rate of convergence
- Attains ε-accuracy within $O(\frac{1}{\mu\lambda} \log \frac{1}{\varepsilon})$ iterations

Proof of Lemma 7

By definition, one has

$$
\begin{aligned}
\|\boldsymbol{x}^{t+1} - \boldsymbol{x}_{\mathsf{opt}}\|_2^2 &= \|\boldsymbol{x}^t - \eta \boldsymbol{g}(\boldsymbol{x}^t) - \boldsymbol{x}_{\mathsf{opt}}\|_2^2 \\
&= \|\boldsymbol{x}^t - \boldsymbol{x}_{\mathsf{opt}}\|_2^2 + \eta^2 \|\boldsymbol{g}(\boldsymbol{x}^t)\|_2^2 - 2\eta \left\langle \boldsymbol{g}(\boldsymbol{x}^t), \boldsymbol{x}^t - \boldsymbol{x}_{\mathsf{opt}} \right\rangle \\
&\le \|\boldsymbol{x}^t - \boldsymbol{x}_{\mathsf{opt}}\|_2^2 + \eta^2 \|\boldsymbol{g}(\boldsymbol{x}^t)\|_2^2 - \eta \left(\lambda \|\boldsymbol{x}^t - \boldsymbol{x}_{\mathsf{opt}}\|_2^2 + \mu \|\boldsymbol{g}(\boldsymbol{x}^t)\|_2^2 \right) \\
&= (1 - \eta\lambda) \|\boldsymbol{x}^t - \boldsymbol{x}_{\mathsf{opt}}\|_2^2 + \eta(\eta - \mu) \|\boldsymbol{g}(\boldsymbol{x}^t)\|_2^2 \\
&\le (1 - \mu\lambda) \|\boldsymbol{x}^t - \boldsymbol{x}_{\mathsf{opt}}\|_2^2
\end{aligned}
$$

Initialization Matters in Nonconvex Problems

- Local convergence results require starting in a basin of attraction
- Good initializations reduce time to reach the fast local regime
- Common strategies:
- spectral initialization
- random initialization + perturbation
- warm start from a convex/relaxed estimator

Limitations and Failure Modes

- Spurious local minima can appear in some models
- Flat/degenerate saddles can slow practical convergence
- Ill-conditioning (large condition number) hurts rates
- Optimization success does not automatically imply statistical generalization

Takeaway: always match algorithms and guarantees to structure in the specific problem.

Part III: Return to the Running Example

Apply the toolkit: local rates and global critical-point geometry

Interesting Questions

$$\text{minimize}_{\boldsymbol{x} \in \mathbb{R}^n} \quad f(\boldsymbol{x}) = \frac{1}{4} \|\boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{M}\|_{\mathrm{F}}^2$$

- How does the curvature behave, at least locally around the global minimizer?
- Where / what are the critical points? (Global geometry)

Gradient and Hessian via $f(\boldsymbol{x}+\boldsymbol{h})$ Expansion

For

$$f(\boldsymbol{x}) = \frac{1}{4}\|\boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{M}\|_{\mathrm{F}}^2, \qquad \boldsymbol{A} := \boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{M},$$

expand

$$f(\boldsymbol{x}+\boldsymbol{h}) = \frac{1}{4}\|\boldsymbol{A} + \boldsymbol{x}\boldsymbol{h}^\top + \boldsymbol{h}\boldsymbol{x}^\top + \boldsymbol{h}\boldsymbol{h}^\top\|_{\mathrm{F}}^2.$$

Collecting terms up to second order in $\boldsymbol{h}$:

$$f(\boldsymbol{x}+\boldsymbol{h}) = f(\boldsymbol{x}) + \underbrace{\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{h}\rangle}_{\text{1st order}} + \frac{1}{2}\underbrace{\boldsymbol{h}^\top\Big(2\boldsymbol{x}\boldsymbol{x}^\top + \|\boldsymbol{x}\|_2^2\boldsymbol{I} - \boldsymbol{M}\Big)\boldsymbol{h}}_{\text{2nd order}} + O(\|\boldsymbol{h}\|_2^3).$$

Hence

$$\nabla f(\boldsymbol{x}) = (\boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{M})\boldsymbol{x}, \qquad \nabla^2 f(\boldsymbol{x}) = 2\boldsymbol{x}\boldsymbol{x}^\top + \|\boldsymbol{x}\|_2^2\boldsymbol{I} - \boldsymbol{M}.$$

Local Linear Convergence of GD

Theorem 8

Suppose that $\|\boldsymbol{x}_0 - \sqrt{\lambda_1}\boldsymbol{u}_1\|_2 \le \frac{\lambda_1-\lambda_2}{15\sqrt{\lambda_1}}$ *and set* $\eta = \frac{1}{4.5\lambda_1}$*, GD obeys*

$$\left\|\boldsymbol{x}^t - \sqrt{\lambda_1}\boldsymbol{u}_1\right\|_2 \le \left(1 - \frac{\lambda_1 - \lambda_2}{18\lambda_1}\right)^t \left\|\boldsymbol{x}^0 - \sqrt{\lambda_1}\boldsymbol{u}_1\right\|_2, \quad t \ge 0,$$

- condition number/eigengap determines rate of convergence
- Requires initialization: use spectral method?

Proof of Theorem 8

It suffices to show that for all $\boldsymbol{x}$ obeying $\underbrace{\|\boldsymbol{x} - \sqrt{\lambda_1}\boldsymbol{u}_1\|_2 \le \frac{\lambda_1 - \lambda_2}{15\sqrt{\lambda_1}}}_{\text{basin of attraction}}$,

$$0.25(\lambda_1 - \lambda_2)\boldsymbol{I}_n \preceq \nabla^2 f(\boldsymbol{x}) \preceq 4.5\lambda_1 \boldsymbol{I}_n$$

Express gradient and Hessian as

$$\begin{aligned}\nabla f(\boldsymbol{x}) &= (\boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{M})\boldsymbol{x} \\ \nabla^2 f(\boldsymbol{x}) &= 2\boldsymbol{x}\boldsymbol{x}^\top + \|\boldsymbol{x}\|_2^2 \boldsymbol{I}_n - \boldsymbol{M}\end{aligned}$$

Preliminary Facts

Let $\boldsymbol{\Delta} := \boldsymbol{x} - \sqrt{\lambda_1}\boldsymbol{u}_1$. It is seen that when $\|\boldsymbol{\Delta}\|_2 \leq \frac{\lambda_1 - \lambda_2}{15\sqrt{\lambda_1}}$, one has

$$\lambda_1 - 0.25(\lambda_1 - \lambda_2) \leq \|\boldsymbol{x}\|_2^2 \leq 1.15\lambda_1;$$
$$\|\boldsymbol{\Delta}\|_2 \leq \|\boldsymbol{x}\|_2;$$
$$\|\boldsymbol{\Delta}\|_2\|\boldsymbol{x}\|_2 \leq (\lambda_1 - \lambda_2)/12$$

Local Smoothness

Triangle inequality gives

$$\begin{aligned}\|\nabla^2 f(\boldsymbol{x})\| &\leq \|2\boldsymbol{x}\boldsymbol{x}^\top\| + \|\boldsymbol{x}\|_2^2 + \|\boldsymbol{M}\| \\ &\leq 3\|\boldsymbol{x}\|_2^2 + \lambda_1 < 4.5\lambda_1,\end{aligned}$$

where the last relation follows from $\|\boldsymbol{x}\|_2^2 \leq 1.15\lambda_1$

Local Strong Convexity

Recall that $\boldsymbol{\Delta} = \boldsymbol{x} - \sqrt{\lambda_1}\boldsymbol{u}_1$

$$\begin{aligned}
\boldsymbol{x}\boldsymbol{x}^\top &= \lambda_1 \boldsymbol{u}_1\boldsymbol{u}_1^\top + \boldsymbol{\Delta}\boldsymbol{x}^\top + \boldsymbol{x}\boldsymbol{\Delta}^\top - \boldsymbol{\Delta}\boldsymbol{\Delta}^\top \\
&\succeq \lambda_1 \boldsymbol{u}_1\boldsymbol{u}_1^\top - 3\|\boldsymbol{\Delta}\|_2\|\boldsymbol{x}\|_2 \boldsymbol{I}_n \qquad (\|\boldsymbol{\Delta}\|_2 \le \|\boldsymbol{x}\|_2) \\
&\succeq \lambda_1 \boldsymbol{u}_1\boldsymbol{u}_1^\top - 0.25(\lambda_1 - \lambda_2)\boldsymbol{I}_n,
\end{aligned}$$

where last line relies on $\|\boldsymbol{\Delta}\|_2\|\boldsymbol{x}\|_2 \le (\lambda_1 - \lambda_2)/12$. Consequently,

$$\begin{aligned}
\nabla^2 f(\boldsymbol{x}) &= 2\boldsymbol{x}\boldsymbol{x}^\top + \|\boldsymbol{x}\|_2^2 \boldsymbol{I}_n - \lambda_1 \boldsymbol{u}_1\boldsymbol{u}_1^\top - \sum\nolimits_{i=2}^n \lambda_i \boldsymbol{u}_i\boldsymbol{u}_i^\top \\
&\succeq 2\lambda_1 \boldsymbol{u}_1\boldsymbol{u}_1^\top + \Big(\|\boldsymbol{x}\|_2^2 - 0.5(\lambda_1 - \lambda_2)\Big)\boldsymbol{I}_n - \lambda_1 \boldsymbol{u}_1\boldsymbol{u}_1^\top - \sum\nolimits_{i=2}^n \lambda_i \boldsymbol{u}_i\boldsymbol{u}_i^\top \\
&\succeq (\|\boldsymbol{x}\|_2^2 - 0.5(\lambda_1 - \lambda_2) + \lambda_1)\boldsymbol{u}_1\boldsymbol{u}_1^\top \\
&\quad + \sum\nolimits_{i=2}^n (\|\boldsymbol{x}\|_2^2 - 0.5(\lambda_1 - \lambda_2) - \lambda_i)\boldsymbol{u}_i\boldsymbol{u}_i^\top \\
&\succeq (\|\boldsymbol{x}\|_2^2 - 0.5(\lambda_1 - \lambda_2) - \lambda_2)\boldsymbol{I}_n \\
&\succeq 0.25(\lambda_1 - \lambda_2)\boldsymbol{I}_n \qquad (\lambda_1 - 0.25(\lambda_1 - \lambda_2) \le \|\boldsymbol{x}\|_2^2)
\end{aligned}$$

Critical Points of $f(\cdot)$

$$\boldsymbol{x} \text{ is a critical point, i.e., } \nabla f(\boldsymbol{x}) = (\boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{M})\boldsymbol{x} = \boldsymbol{0}$$

$$\Updownarrow$$

$$\boldsymbol{M}\boldsymbol{x} = \|\boldsymbol{x}\|_2^2 \boldsymbol{x}$$

$$\Updownarrow$$

$$\boldsymbol{x} \text{ aligns with an eigenvector of } \boldsymbol{M} \quad \text{or} \quad \boldsymbol{x} = \boldsymbol{0}$$

Since $\boldsymbol{M}\boldsymbol{u}_i = \lambda_i \boldsymbol{u}_i$, the set of critical points is given by

$$\{\boldsymbol{0}\} \cup \{\pm\sqrt{\lambda_i}\boldsymbol{u}_i, \quad i = 1, \ldots, n\}$$

Categorization of Critical Points

The critical points can be further categorized based on the **Hessian**:

$$\nabla^2 f(\boldsymbol{x}) = 2\boldsymbol{x}\boldsymbol{x}^\top + \|\boldsymbol{x}\|_2^2 \boldsymbol{I}_n - \boldsymbol{M}$$

- For any non-zero critical point $\boldsymbol{x}_k = \pm\sqrt{\lambda_k}\boldsymbol{u}_k$:

$$\begin{aligned}
\nabla^2 f(\boldsymbol{x}_k) &= 2\lambda_k \boldsymbol{u}_k \boldsymbol{u}_k^\top + \lambda_k \boldsymbol{I} - \boldsymbol{M} \\
&= 2\lambda_k \boldsymbol{u}_k \boldsymbol{u}_k^\top + \lambda_k \left(\sum_{i=1}^n \boldsymbol{u}_i \boldsymbol{u}_i^\top \right) - \sum_{i=1}^n \lambda_i \boldsymbol{u}_i \boldsymbol{u}_i^\top \\
&= \sum_{i:i\neq k} (\lambda_k - \lambda_i) \boldsymbol{u}_i \boldsymbol{u}_i^\top + 2\lambda_k \boldsymbol{u}_k \boldsymbol{u}_k^\top
\end{aligned}$$

Categorization of Critical Points (Cont.)

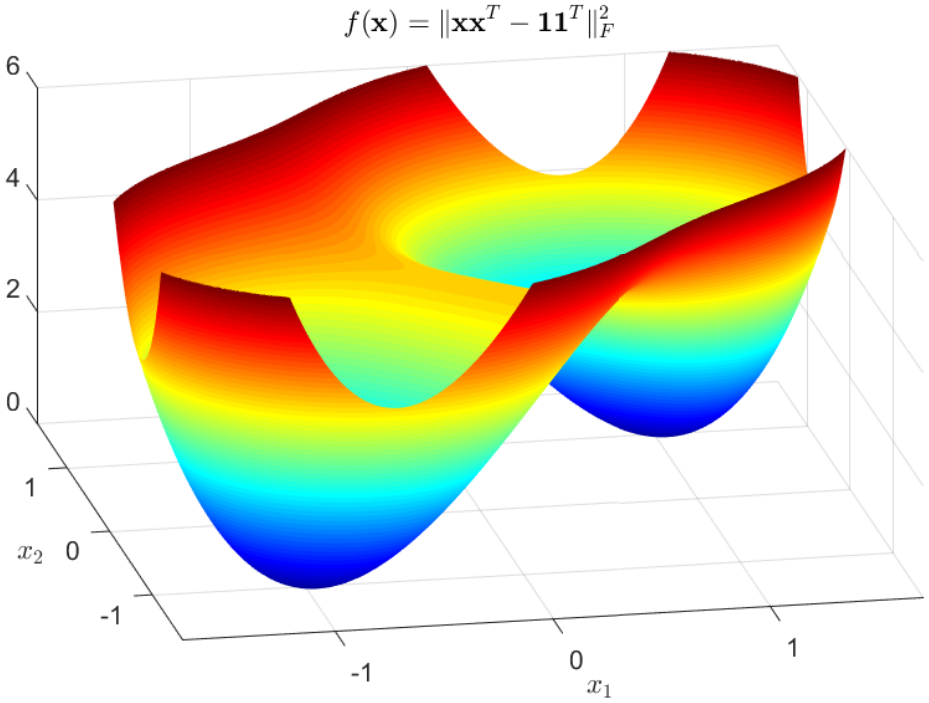
If $\lambda_1 > \lambda_2 \geq \ldots \geq \lambda_n \geq 0$, then

- $\nabla^2 f(\boldsymbol{x}_1) \succ \mathbf{0}$ $\quad\rightarrow\quad$ local minima
- $1 < k \leq n$: $\lambda_{\min}(\nabla^2 f(\boldsymbol{x}_k)) < 0$, $\lambda_{\max}(\nabla^2 f(\boldsymbol{x}_k)) > 0$ $\quad\rightarrow\quad$ strict saddle
- $\boldsymbol{x} = \mathbf{0}$: $\nabla^2 f(\mathbf{0}) = -\boldsymbol{M} \preceq \mathbf{0}$ $\quad\rightarrow\quad$ local maxima, strict saddle

all local minima are global; all saddles are strict

A Pictorial Example

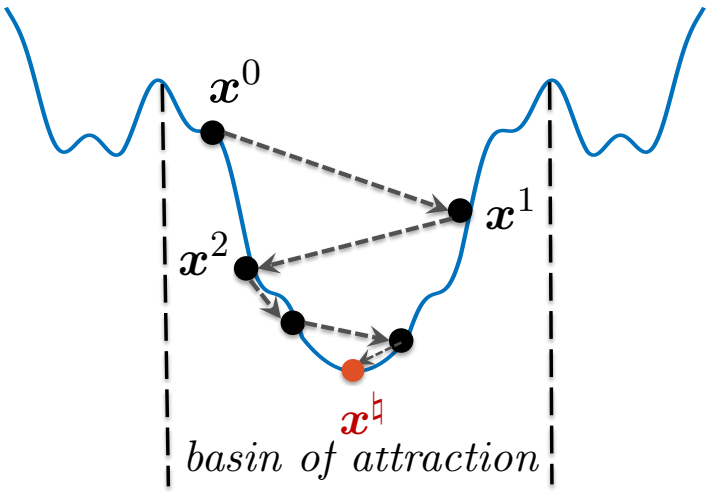
For example, for 2-dimensional case $f(\boldsymbol{x}) = \left\| \boldsymbol{x}\boldsymbol{x}^\top - \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right\|_{\mathrm{F}}^2$



global minima: $\boldsymbol{x} = \pm \begin{bmatrix} 1 \\ 1 \end{bmatrix}$; strict saddles: $\boldsymbol{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, and $\pm \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

— No "spurious" local minima!

Two-stage approach:



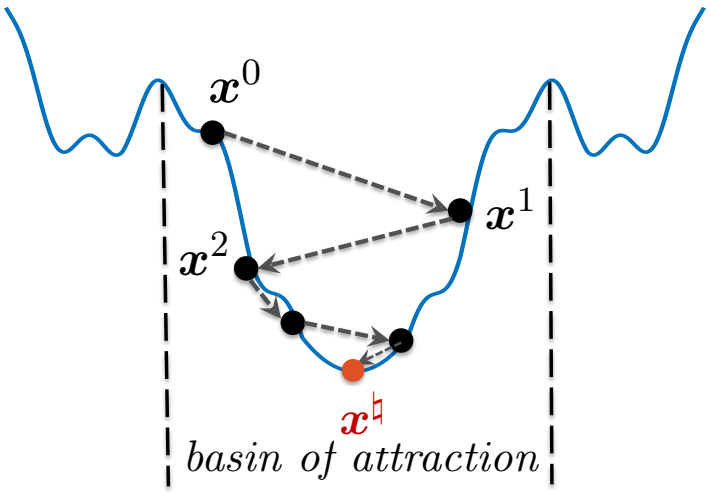
smart initialization

$+$

local refinement

Two Vignettes

Two-stage approach:

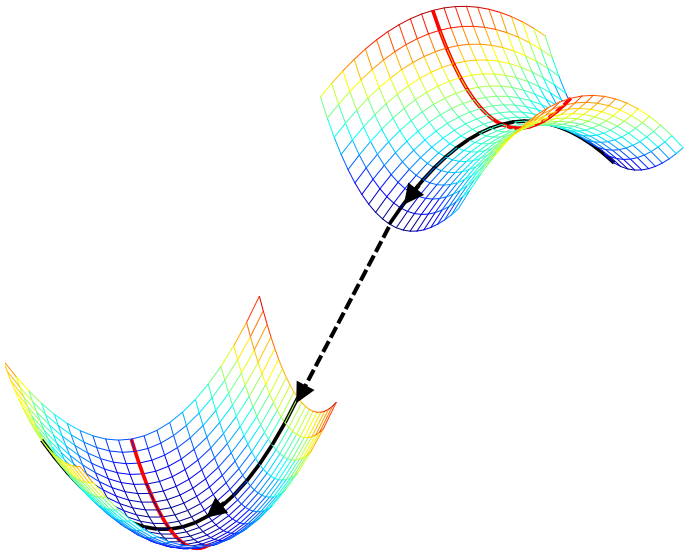


smart initialization

$+$

local refinement

Global landscape:



benign landscape

$+$

saddle-point escaping

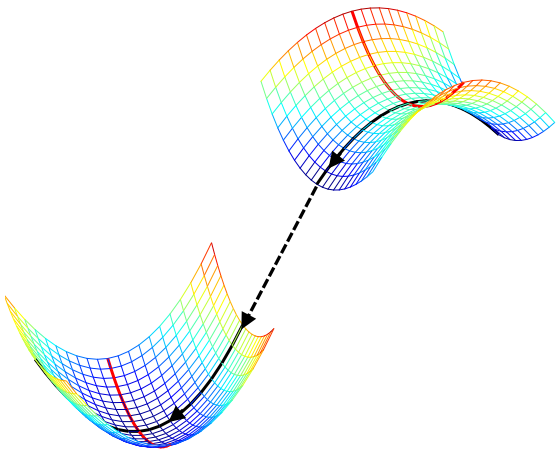
Global Landscape

Benign landscape:

- all local minima = global minima
- other critical points = strict saddle points

Saddle-point escaping algorithms:

- trust-region methods
- perturbed gradient descent
- perturbed SGD
- ...



Next Steps

- Generic local analysis of (regularized) gradient descent
- Refined local analysis for gradient descent
- Global landscape analysis
- Gradient descent with random initialization
- (Maybe) Gradient descent with arbitrary initialization